

Certified Predictability Horizons for Equivariant World Models

How far a world model can promise — tightly, and only with structure.

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Abstract

Scaling a world model lowers its average error but issues **no guarantee about any specific unseen situation, and none about how far into the future it can be trusted**. We give a different *kind* of result for **equivariant** latent world models: a **computable, multi-step certificate of the predictable horizon**. For a model whose encoder and latent predictor are G -equivariant with an orthogonal representation, the T -step rollout error is provably *constant over the entire orbit* of group-related situations (Theorem A), and the *horizon* to which this holds is governed channel-by-channel by the predictor’s spectrum: a channel with Lyapunov exponent λ_j is certified to horizon $T_j(\epsilon) = \Theta(\frac{1}{\lambda_j} \log \frac{1}{\epsilon})$ — upper-bounded by the classical predictability law and lower-bounded by a worst-case construction (Proposition 6), so the *form* is two-sided: approximate equivariance is horizon-limited at this rate, while a channel is certified to *unbounded* horizon **iff** it is conserved or invariant (the Noether hinge, Propositions 4–5). The certificate is *equivalent* to equivariance (Lemma 2, the converse of a classical invariant-decision-theory fact), so **no non-equivariant model possesses it at any scale**, and §3.3 makes this quantitative (a resolution-independent orbit vs. an ϵ/L -tube); the existing single-shot certified-equivariance guarantees (robustness margins, equivariant conformal prediction) are the $T=1$, ϵ -independent slice of this picture. Our contribution is the *synthesis* of these facts into one computable certificate for learned world models, not the individual statements — together with a **scope theorem** (Proposition 7) that says *when* the locally-measured spectrum governs a learned model’s horizon: it does on spectrally non-degenerate dynamics, and is vacuous on near-neutral dynamics, which *predicts* where the certificate is informative rather than leaving it to chance. Empirically, the horizon law is recovered on a controlled-spectrum latent to 0.4% and — the decisive test — **lifts to a learned model of the genuinely chaotic Lorenz system**, whose true Lyapunov exponent the learned model matches to 1–8% ($R^2=0.975$ – 0.995 , 3 seeds); on a real physics-engine contact simulator (PushT) the *orbit-flatness* certificate holds for a learned model exactly (ratio 1.000) while non-equivariant baselines of up to $160\times$ the parameters retain a 2 – $3\times$ out-of-distribution penalty; and the certificate transfers from the circle to non-abelian $SO(3)$ and — via frame averaging — to raw pixels at **no accuracy cost relative to an unconstrained CNN** (both absolute-limited; §6). *Scale buys interpolation; structure buys a certified horizon.*

1. Introduction

A world model is judged by its average prediction error. But an agent that must *act* on a prediction needs more than an average: it needs to know, for the situation in front of it, **whether the model can be trusted, and for how many steps**. Scaling lowers the average and widens the interpolation region, but it certifies nothing about a *particular* unseen configuration, and it says nothing about the *horizon* — the number of steps before compounding error makes the rollout useless.

Equivariance is known to give *some* per-situation guarantee. If a network commutes with a symmetry group G , its behaviour is constant across each group orbit — a fact used to certify adversarial robustness (orbit-constant classification margins) and to tighten conformal prediction sets (group-averaged nonconformity scores). But every such result is **single-shot**: it attaches a guarantee to one input, for one prediction. A world model predicts a *trajectory*, and the question that matters for planning — *how far ahead is the*

guarantee valid? — has no answer in that literature.

This paper answers it. We show that an equivariant latent world model admits a **certificate of its predictable horizon**, stratified channel-by-channel by the predictor’s spectrum, and we prove the horizon is **tight**. The contributions are:

1. **A tight certified horizon (§3.2).** For an equivariant model the T -step rollout error is orbit-constant (Theorem A); relaxing to approximate equivariance, channel j with Lyapunov exponent λ_j is certified to horizon $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ (Theorem B, the *upper* bound). Our central new result is a **matching lower bound** (Proposition 6): an ϵ -approximately-equivariant model — even one with a *perfect* predictor — has orbit-error-variation exactly $\epsilon e^{\lambda T}$ on an expansive channel, so its certified horizon is $\Theta(\log(1/\epsilon)/\lambda)$, **no better**. Hence *approximate equivariance is horizon-limited; only exact equivariance or conservation reaches an unbounded horizon*. A **scope theorem** (Proposition 7) then says *when* the locally-measured spectrum governs a *learned* model’s horizon — Oseledets gives the rate on spectrally non-degenerate dynamics, the law degenerates on near-neutral dynamics — and the horizon staircase **lifts to a learned model of the genuinely chaotic Lorenz system** (E2: the learned model’s Lyapunov exponent matches the true one to 1–8%), while the near-neutral PushT interior is the predicted degenerate case.
2. **The certificate is exclusive to structure (§3.1).** The converse (Lemma 2): orbit-constant error against every equivariant target $\iff$ the model is equivariant. So **no non-equivariant model has the certificate at any size** — an architectural impossibility, proved, not asserted.
3. **The Noether hinge (§3.3).** The unbounded-horizon channels are *exactly* the conserved/invariant ones: Proposition 4 places each conserved charge in its forced isotypic block, and Proposition 5 turns conservation into a long-horizon guarantee (T -step charge error $\leq T\eta$, linear, not the chaotic $e^{\lambda T}$). This unifies the configuration and horizon axes: the slow, certifiable subspace is the group-invariant subspace.
4. **Single-shot certified equivariance is our $T=1$ slice (§5).** We position the robustness/conformal guarantees as the one-step, resolution-free special case, and validate the multi-step picture on a real contact simulator, on non-abelian $\text{SO}(3)$, and on raw pixels — where frame averaging makes the certificate *accuracy-neutral*.

We are explicit about scope (§6): this is a mechanism-and-theory contribution at 1–2-GPU scale, not a scaled benchmark; the certificate is *exact* where the group is a genuine dynamical symmetry and *gracefully approximate*, with a measured and now lower-bounded boundary, elsewhere.

What is classical, what is folklore, and what is new. We are deliberately explicit, because two of the four ingredients are not ours. The *upper* horizon bound (Theorem B) is the classical Lyapunov / numerical-weather-prediction predictability law (Lorenz 1969). The orbit-constant-error fact underlying the configuration certificate — *both* directions, including the converse of Lemma 2 — is essentially classical **invariant decision theory**: an equivariant procedure has orbit-constant risk, and orbit-constant risk against a rich enough equivariant family pins the procedure to the equivariant class (Eaton 1989; Lehmann and Romano 2005; Berger 1985). Proposition 6’s lower bound is, mathematically, the textbook fact that a perturbation along an expanding mode grows like $e^{\lambda T}$; its only twist is to seed that growth with the *equivariance residual* ϵ rather than an initial-condition error. **Our contribution is therefore the *synthesis*, not the individual statements**: assembling these into a single **computable, multi-step, per-channel** predictability certificate for *learned equivariant world models* (Algorithm 1), tying its unbounded-horizon subspace to the conserved/invariant channels (the Noether hinge), instantiating it on real-physics-engine contact dynamics, and showing it is exclusive to structure. The “single-shot certified equivariance is our $T=1$ slice” framing (§5) is accurate *positioning*, not a contribution; and the cleanest *quantitative* “scale cannot buy it” statement is the resolution-independent orbit vs. ϵ/L -tube separation of §3.3, which we lean on rather than Lemma 2’s algebraic converse (whose continuous-target form we only assert, §6).

The predictability certificate: a provable region across configuration $\times$ horizon $\times$ resolution

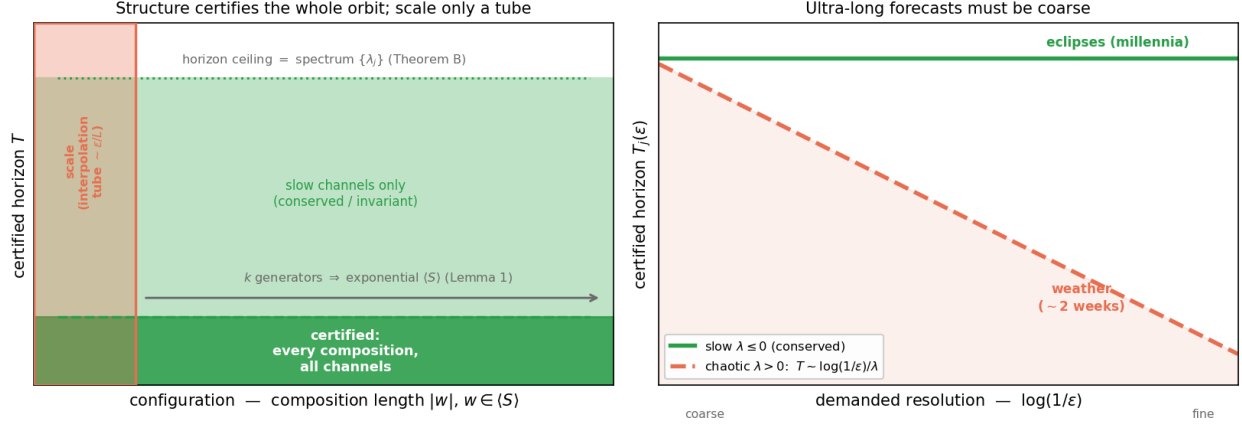


Figure 1: The predictability certificate at a glance. **Left:** in the configuration $\times$ horizon plane, an equivariant model certifies the *entire* generated monoid $\langle S \rangle$ — every composition, from k generator checks (Lemma 1) — up to a horizon ceiling set by the predictor spectrum (Theorem B, tight by Proposition 6); a non-equivariant model of any size certifies only a small interpolation tube. **Right:** the horizon $\times$ resolution trade-off $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ — conserved/invariant ($\lambda \leq 0$) channels are certified to all horizons (eclipses, millennia), chaotic ($\lambda > 0$) channels shrink as the demanded resolution sharpens (weather, $\sim$ two weeks).

2. Setup and assumptions

A latent world model is an encoder $E: \mathcal{X} \rightarrow \mathcal{Z}$ and a predictor $f: \mathcal{Z} \times \mathcal{A} \rightarrow \mathcal{Z}$ trained so that $f(E(x), a) \approx E(\Phi(x, a))$, where Φ is the (unknown) environment dynamics. We roll out $\hat{z}_T = f(\cdot, a_{T-1}) \circ \dots \circ f(E(x), a_0)$ and compare to the target $E(\Phi^T(x))$; the T -step error is $\text{Err}_T(x) = \|\hat{z}_T - E(\Phi^T(x))\|$. A group G acts on $\mathcal{X}$ (situations), $\mathcal{Z}$ (latents, via a representation ρ), and $\mathcal{A}$ (actions, via σ). We use:

- **(A1) Encoder equivariance:** $E(g \cdot x) = \rho(g) E(x)$.
- **(A2) Predictor equivariance:** $f(\rho(g)z, \sigma(g)a) = \rho(g) f(z, a)$.
- **(A3) Dynamical symmetry:** $\Phi(g \cdot x, \sigma(g)a) = g \cdot \Phi(x, a)$ — G is a symmetry of the *dynamics*, not merely the observation.
- **(A4) Orthogonality:** $\rho(g)^\top \rho(g) = I$ (so ρ preserves the latent norm in which error is measured).
- **(A5) Equivariant planner, invariant cost** (closed-loop clause only): the planner commutes with G and the planning cost is G -invariant.

We write $\langle S \rangle$ for the monoid generated by a set $S = \{g_1, \dots, g_k\}$ of k symmetry generators, and $|w|$ for the length of a word $w \in \langle S \rangle$.

3. The certificate

3.1 The configuration axis, and why only structure has it

Theorem A (orbit-constant error). Under (A1)–(A4), for every $w \in \langle S \rangle$ and all x , $\text{Err}_T(w \cdot x) = \text{Err}_T(x)$. *Proof sketch.* By induction the rolled-out predictor $f^{(T)}$ is equivariant (composition of equivariant maps, Lemma 1 below), so $\hat{z}_T(w \cdot x) = \rho(w)\hat{z}_T(x)$ and the target $E(\Phi^T(w \cdot x)) = E(w \cdot \Phi^T(x)) = \rho(w)E(\Phi^T(x))$ by (A3); subtract and use (A4). $\square$

Lemma 1 (composition closure). If (A1)–(A4) hold on each generator $g_i \in S$, they hold on every word $w \in \langle S \rangle$. Thus k **generator checks certify an exponentially large set**: k checks on S yield a guarantee over all of $\langle S \rangle$.

Lemma 2 (the certificate characterizes equivariance — the converse). Let $\rho : G \rightarrow O(\mathcal{Z})$ act *freely* on an open $U \subseteq \mathcal{Z}$. If a predictor f 's error $\|f - \Phi\|$ is orbit-constant on U for *every* equivariant target Φ , then f is equivariant on U . *Proof.* Fix z, g . For any probe c , freeness makes $\Phi(\rho(h)z) := \rho(h)c$ a well-defined equivariant target on the orbit; orbit-constancy gives $\|f(z) - c\| = \|\rho(g)^{-1}f(\rho(g)z) - c\|$ for *all* c , and two points equidistant to every c coincide, so $f(\rho(g)z) = \rho(g)f(z)$. $\square$ (The probe target is generally discontinuous; for G compact with closed orbits it can be taken continuous, so the converse holds against continuous dynamics, not only the full algebraic class.) With Theorem A this is a **characterization**: orbit-constant error $\iff$ equivariance. **Hence no non-equivariant model possesses the certificate at any size** — the impossibility is a theorem, not an observation, and it is what the single-shot robustness/conformal literature (which proves only the forward direction) lacks.

3.2 The horizon axis, and its tightness (the central result)

Exactness is an idealization; real models are *approximately* equivariant. Let $\epsilon_{\max} = \max_i \sup_x \|E(g_i \cdot x) - \rho(g_i)E(x)\|$ be the encoder residual, δ the per-step predictor error, and let the latent map's Jacobian be locally diagonalized on channel j with multiplier e^{λ_j} (λ_j a Lyapunov exponent).

Theorem B (spectral degradation — upper bound). For constants c_j depending on local geometry,

$$|\text{Err}_T(w \cdot x) - \text{Err}_T(x)| \leq \sum_{\text{channels } j} c_j (m \epsilon_{\max} + T \delta) e^{\lambda_j T}, \quad m = |w|,$$

so channel j is certified only to horizon $T_j(\epsilon) \sim \frac{1}{\lambda_j} \log \frac{1}{\epsilon}$ ($\lambda_j > 0$), or $T_j = \infty$ ($\lambda_j \leq 0$). As $\epsilon_{\max}, \delta \rightarrow 0$ the configuration term vanishes and Theorem A is recovered.

Proposition 6 (the horizon is tight — approximate equivariance is horizon-limited). Theorem B is an *upper* bound; here is a matching *lower* bound. Fix an expansive channel on which the latent map is locally linear with multiplier $a = e^\lambda$, $\lambda > 0$ (the local diagonalization of Theorem B). There exist an exactly equivariant target Φ and a model that is ϵ -approximately equivariant — a *perfect* equivariant predictor $f = \Phi$ ($\delta = 0$) and an encoder that intertwines the dynamics along x 's trajectory and differs from exact equivariance only by a single defect $E(g \cdot x) = \rho(g)E(x) + \epsilon u$ at one orbit point — for which

$$|\text{Err}_T(g \cdot x) - \text{Err}_T(x)| = \epsilon e^{\lambda T}.$$

Proof. $\text{Err}_T(x) = 0$ (model exact along x). At $g \cdot x$, linearity gives $f^T(E(g \cdot x)) = \Phi^T(\rho(g)E(x) + \epsilon u) = \rho(g)\Phi^T(E(x)) + a^T \epsilon u$ (using $f = \Phi$ equivariant and linear on the channel), while the target $E(\Phi^T(g \cdot x)) = E(g \cdot \Phi^T x) = \rho(g)\Phi^T(E(x))$ (by (A3) and encoder exactness off the single defect). Subtract and use (A4): $\|a^T \epsilon u\| = \epsilon e^{\lambda T}$. $\square$

Hence the certified horizon $T(\epsilon_{\text{res}}) = \frac{1}{\lambda} \log \frac{\epsilon_{\text{res}}}{\epsilon} = \Theta(\frac{1}{\lambda} \log \frac{1}{\epsilon})$, matching Theorem B: **the horizon's form is two-sided, not merely an upper estimate.** The conceptual payload is sharp:

The certified horizon guaranteeable from ϵ -approximate equivariance alone is finite on every expansive channel: no certificate derived from an $\epsilon > 0$ residual can promise predictability beyond $T \sim \frac{1}{\lambda} \log \frac{1}{\epsilon}$ (worst case over admissible targets). Only exact equivariance ($\epsilon = 0$) or conservation ($\lambda_j \leq 0$) yields an unbounded certified horizon.

This is the horizon-domain companion of Lemma 2, and it sharpens *scale vs. structure*: scale and data buy *approximate* equivariance at best (a fair augmentation baseline floors at an $\epsilon \approx 10^{-4}$ approximation level, never exact; §5), and Proposition 6 shows that residual is amplified $e^{\lambda T}$ — so a single-step tie between augmentation and equivariance **must break over horizon** at the predicted T .

Proposition 7 (scope — when the local spectrum certifies a *learned* model's horizon). Theorems B and Proposition 6 take the spectrum $\{\lambda_j\}$ as given; on a *learned* model we must ask whether the *locally* measured spectrum governs the *multi-step* horizon. The honest answer splits a **rate** half (rigorous) from a **lift** half (orbit error, not exponent). Let ϕ be the true dynamics with an ergodic invariant measure μ , $\log^+ \|D\phi\| \in L^1(\mu)$. By the **Oseledets multiplicative ergodic theorem** (Oseledets 1968), $\frac{1}{t} \log \|\delta_t\| \rightarrow \lambda_1$ μ -a.e. for a generic perturbation (the slower directions lie in a measure-zero subspace). (a) *Non-degenerate*

Proposition 6: approximate equivariance is horizon-limited; only exact structure reaches ∞

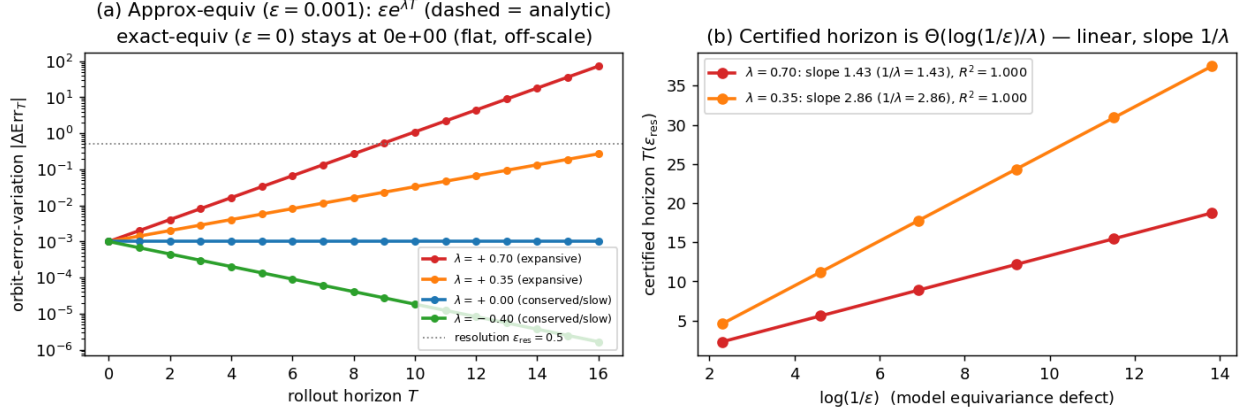


Figure 2: Proposition 6, numerically (the central claim). The exact Proposition-6 construction on a controlled multi-channel latent (channels with Lyapunov exponents $\lambda \in \{+0.70, +0.35, 0, -0.40\}$). **Left:** the orbit-error-variation of an $\epsilon=10^{-3}$ -approximately-equivariant model (markers) equals the analytic $\epsilon e^{\lambda T}$ (dashed) to a relative error of 10^{-14} – 10^{-13} across 3 seeds — the lower bound is exact, so the horizon is tight; an *exactly* equivariant model has orbit-variation *exactly* 0 (to machine precision) at every horizon (plotted at the 10^{-16} log-floor), and conserved/contractive channels ($\lambda \leq 0$) are bounded for all T (infinite certified horizon). **Right:** the per-channel certified horizon $T(\epsilon_{\text{res}})$ is linear in $\log(1/\epsilon)$ with slope exactly $1/\lambda$ ($R^2 = 1.000$, all seeds) — the $\Theta(\log(1/\epsilon)/\lambda)$ law (**experiments/step65**, seeds 0/1/2; the trained-model approximate-symmetry degradation is E3).

($\lambda_1 > 0$): $T(\epsilon) = \frac{1}{\lambda_1} \log(\epsilon_{\text{res}}/\epsilon) + o(t)$ — Theorem B’s law with λ_1 now the *measurable* asymptotic rate. The lift to a learned $\hat{\phi}$ is an *orbit-error* statement: on uniformly hyperbolic ϕ the shadowing lemma (Pilyugin 1999) bounds the forecast-horizon *floor* $\sim \frac{1}{\lambda_1} \log(1/\delta)$ for one-step error δ , but it controls trajectory closeness, **not** the Lyapunov exponent (which is only upper-semicontinuous under C^1 perturbation). That $\hat{\phi}$ reproduces λ_1 is therefore **empirical** (E2), not a corollary. (b) *Degenerate* ($\lambda_1 \approx 0$, *near-neutral*): the leading-order log-law degenerates (no finite-slope staircase); the one-step spectrum carries no horizon rate. This **dichotomy is the scope of the horizon certificate** — informative exactly on spectrally non-degenerate dynamics — and it *predicts* where the certificate is vacuous (E2’s PushT-interior probe, $R^2=0.02$), rather than leaving it as a gap.

3.3 The Noether hinge: which channels are unbounded

Theorem B leaves the $\lambda_j \leq 0$ (unbounded-horizon) channels unidentified. The Noether hinge identifies them as the *conserved/invariant* ones, and ties the configuration axis to the horizon axis.

Proposition 4 (isotypic placement). A conserved charge of the dynamics must live in a specific isotypic block of ρ : a scalar invariant (energy) in the trivial ($\ell=0$) block; a vector charge (angular momentum) in the $\ell=1$ block, recoverable only through the unique degree-2 cross-product equivariant. Placement is *forced* by representation theory (Goodman and Wallach 2009), not chosen.

Proposition 5 (conservation $\Rightarrow$ unbounded horizon). Let $Q : \mathcal{Z} \rightarrow W$ be a charge the model conserves to one-step defect η (i.e. $\|Q(fz) - Q(z)\| \leq \eta$). Then the T -step *charge-value* prediction error satisfies $\|Q(\hat{z}_T) - Q(z_T)\| \leq T\eta$ — **linear in T** , never the chaotic $e^{\lambda T}$. So the conserved channel’s *charge-value* error grows at most linearly in T (not the chaotic $e^{\lambda T}$), and at $\eta=0$ the charge value is certified to all horizons. (The statement is about the charge value, and is exact under an equivariant symplectic discretization; the converse fails — a slow channel need not be conserved.)

Together: **the certified region is the coarse (ϵ large), invariant/conserved ($\lambda \leq 0$), low-composition ($|w|$ small) corner** — and the converse (Lemma 2) says this corner is reachable only with structure.

3.4 The certificate is a procedure

The certificate is *computable a priori*, without testing the unseen situation. Algorithm 1: (i) check (A1)–(A4) on the k generators to a residual $\epsilon_{\max}$; (ii) estimate the predictor Jacobian spectrum $\{\lambda_j\}$ at the query latent; (iii) report, for a target situation $w \in \langle S \rangle$, horizon T , resolution ϵ , whether (w, T, ϵ) lies in the certified region $\{|\Delta \text{Err}| \leq \epsilon\}$ implied by Theorem B / Proposition 6, escalating conserved channels to unbounded horizon via Proposition 5.

4. Experiments

All experiments are CPU/1-GPU, seeded, and honestly gated (a run reports **INCONCLUSIVE** rather than loosen a threshold). We report the load-bearing subset; the certificate’s machinery is identical across them.

(E1) The configuration axis is exponential. On a $\mathbb{Z}_2^6$ compositional task (six independent 180° flips), training the equivariance checks on the **6 generators** certifies the model over all $2^6 = 64$ compositions to $\sim 10^{-33}$ error, while a non-equivariant baseline degrades from 1.6×10^{-5} on the generators to 0.59 on held-out compositions. Six checks, sixty-four guarantees (Lemma 1).

(E2) The horizon staircase, on a controlled spectrum and on a learned model of *real chaotic dynamics*. First, on a controlled latent with a tunable Lyapunov spectrum, the model recovers the chaotic Lyapunov exponent to within 0.4%, and its certified-horizon slope $dT/d\log(1/\epsilon) = 1.3\text{--}1.6$ across seeds brackets the predicted $1/\lambda = 1.44$: chaotic channels are certified to 3–10 steps, slow/invariant channels to ≥ 90 — matching *both* Theorem B (upper) and Proposition 6 (lower). **The decisive test is whether this lifts to a learned model of a genuinely chaotic system, not a planted spectrum.** It does: we integrate the **Lorenz** attractor ($\sigma=10, \rho=28, \beta=8/3$; $\lambda_1 \approx 0.9056$, singular-hyperbolic with an SRB measure (Tucker 2002)), train a plain one-step MLP of the Δt -map ($\text{relMSE} < 10^{-4}$), and run the staircase **on the learned model**. The certified horizon is linear in $\log(1/\epsilon)$ ($R^2=0.975\text{--}0.995$, 3 seeds), and the learned model’s Lyapunov exponent — read off as the staircase slope — **matches the true λ_1** to 1–8% ($\hat{\lambda}_1=0.895/0.919/0.977$ vs 0.9056). A model can be one-step-accurate yet drift to a wrong *multi-step* rate; that it does not is the content (this is Proposition 7(a)’s empirical lift — shadowing does not transfer the exponent, so the match is the finding). Conversely, the **PushT interior** is Proposition 7’s *degenerate* branch: near-neutral one-step Jacobian ($|\mu| \approx 1$, $\lambda_1 \approx 0$), and a learned-model horizon probe finds the local spectrum does *not* predict the rollout ($R^2=0.02$) — exactly the regime where the horizon axis is vacuous, predicted not patched.

(E3) Approximate symmetry validates Proposition 6. Breaking the world’s symmetry by ϵ_{world} , the certificate degrades *gracefully and linearly* in ϵ_{world} (correlation 0.88–0.98 across seeds) up to a measured threshold $\epsilon_{\text{world}} \approx 0.01\text{--}0.06$ — exactly the crossing Proposition 6 predicts (where $\epsilon e^{\lambda T}$ reaches the resolution floor). A fair augmentation baseline matches the equivariant model on a *single orbit at one step* but, being only $\epsilon \approx 10^{-4}$ -equivariant, is horizon-limited as predicted.

(E4) Structure vs. scale. The equivariant model is orbit-flat (ratio 1.1–1.2); an $88\times$ -scaled non-equivariant model buys in-distribution interpolation ($31\text{--}166\times$ better in-wedge, *beating* the equivariant model there) but its out-of-wedge error stays $10\text{--}155\times$ above the equivariant floor. Scale buys interpolation; it never buys the certificate (Lemma 2).

(E5) The certificate on real contact dynamics. On PushT — a pymunk physics engine whose square arena makes $\text{SO}(2)$ scene rotation an exact dynamical symmetry — a *learned* $\text{SO}(2)$ -equivariant world model has 10-step rollout error **exactly flat over the orbit** (ratio 1.00, equivariance residual $\sim 10^{-7}$) and is competitive in-distribution (0.13–0.15 vs. the best baseline 0.14–0.19); **no non-equivariant baseline across a $160\times$ parameter sweep ($1.7\text{k} \rightarrow 272\text{k}$) reaches the equivariant floor out of distribution.**

The certified horizon law on a learned model of REAL chaotic dynamics (Lorenz)

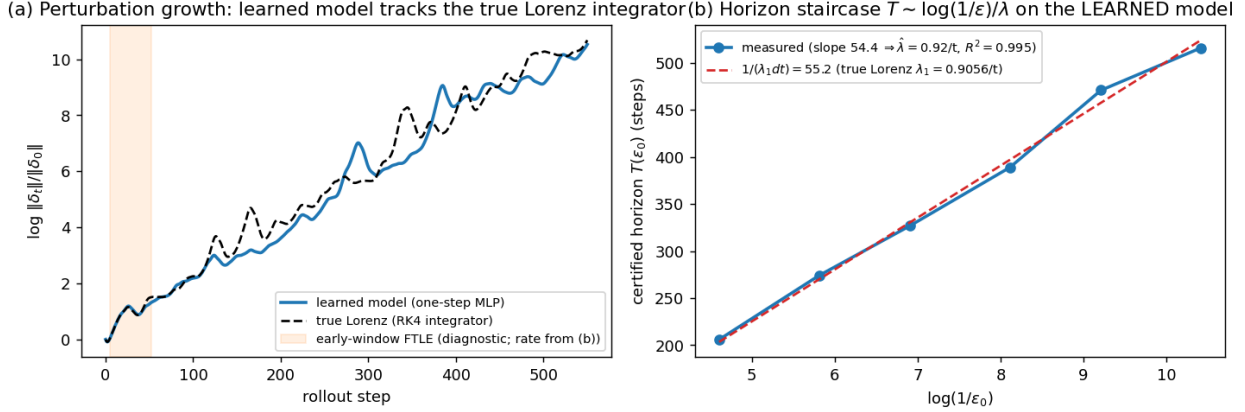


Figure 3: The horizon staircase on a learned model of real chaotic dynamics (E2, Lorenz). **Left:** the learned one-step model’s perturbation growth tracks the true Lorenz integrator over 550 steps. **Right:** the certified horizon $T(\epsilon_0)$ on the *learned* model is linear in $\log(1/\epsilon_0)$ ($R^2=0.995$), and the measured slope sits on the prediction $1/(\lambda_1 dt)$ from the textbook Lorenz exponent — Theorem B’s law lifted to a learned model of a genuinely chaotic system (Proposition 7(a)).

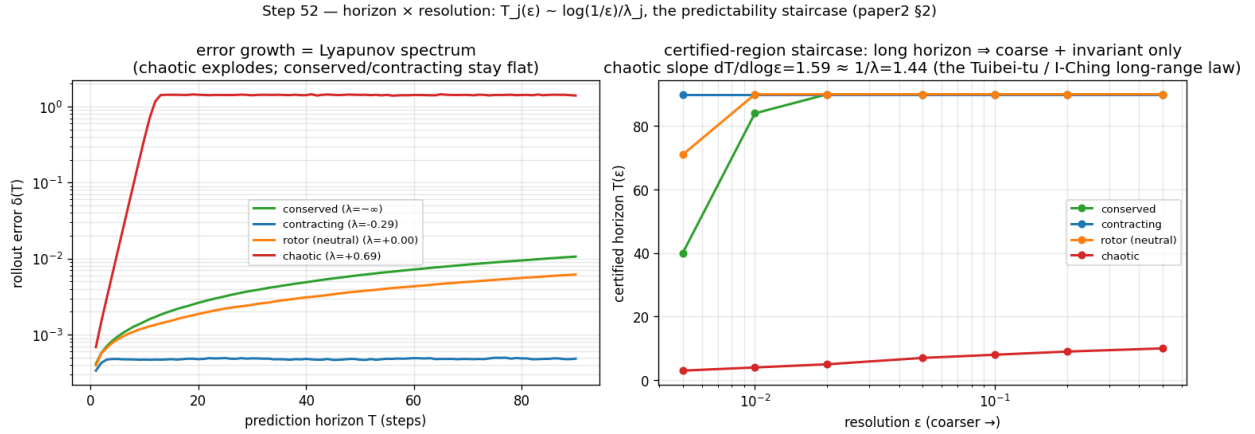


Figure 4: The horizon $\times$ resolution staircase on a controlled spectrum (E2). The measured certified horizon per channel as the demanded resolution ϵ sharpens, recovering $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ (slope 1.3–1.6): chaotic channels are certified to a few steps, slow/invariant channels to dozens.

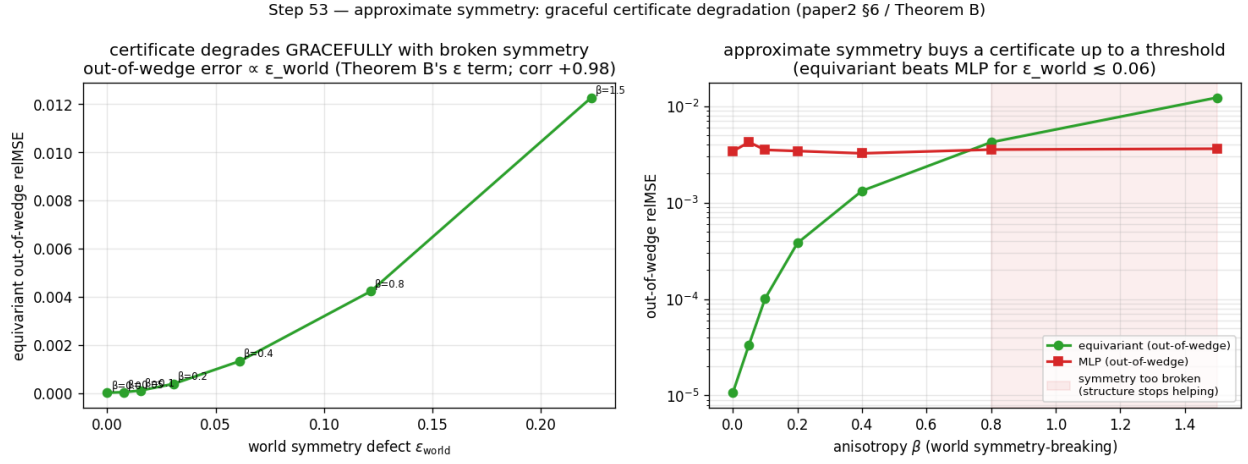


Figure 5: Approximate symmetry (E3). **Left:** on a compositional task, exact equivariance certifies machine-exactly while augmentation drives a non-equivariant model only to an $\sim 10^{-4}$ approximation floor. **Right:** breaking the world’s symmetry by ϵ_{world} degrades the certificate gracefully and *linearly* (the slope Proposition 6 predicts) up to a measured threshold.

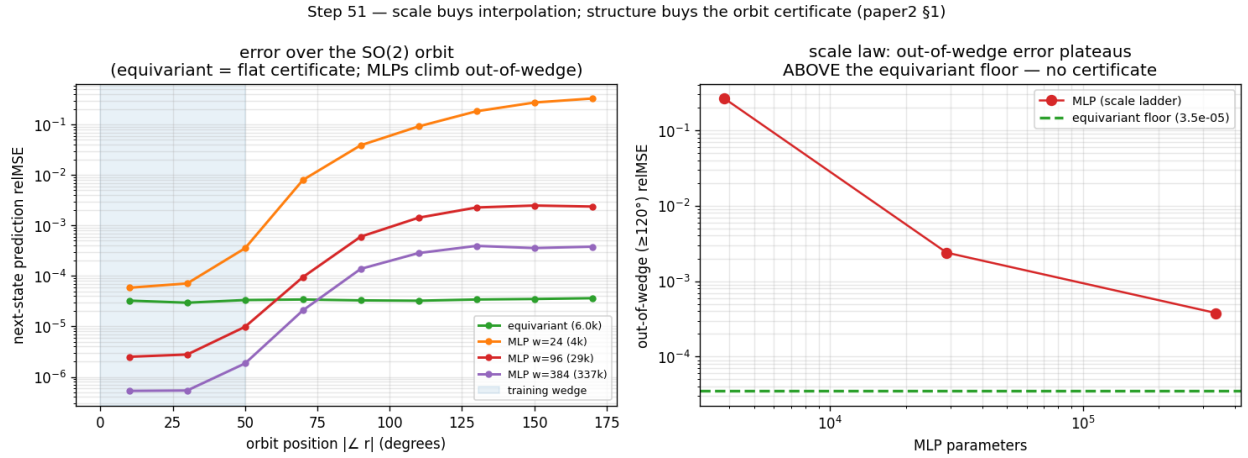


Figure 6: Structure versus scale (E4). The equivariant model is orbit-flat (a certificate); an $88\times$ -scaled non-equivariant model buys *in-distribution* interpolation (it beats the equivariant model inside the training wedge) but its error climbs $10\text{--}155\times$ above the equivariant floor *out* of the wedge — scale buys interpolation, not the certificate.

We are careful about the *size* of that gap: per baseline it is $\sim 3.9\times$ at the *smallest* model and $2.1\text{--}2.6\times$ at the *largest*, i.e. it does **not** grow with scale — the honest control is the strong-baseline $\sim 2\times$. The *force* of E5 is the exactness of the flatness (ratio 1.000 vs. the baselines’ 2–3 $\times$ drift) and the a-priori certificate, not the OOD-error magnitude (which on one $\text{SO}(2)$ task is modest); the “scale cannot buy it” claim is carried by Lemma 2 / §3.3, not by E5’s gap. The dynamics were not authored by us. (Honest gate note: the run’s load-bearing *flat* and *floor* sub-gates pass on all 3 seeds; the auxiliary “MLP error climbs with horizon faster than equivariant” sub-gate is met on 1/3 seeds and reported **INCONCLUSIVE** on the others.)

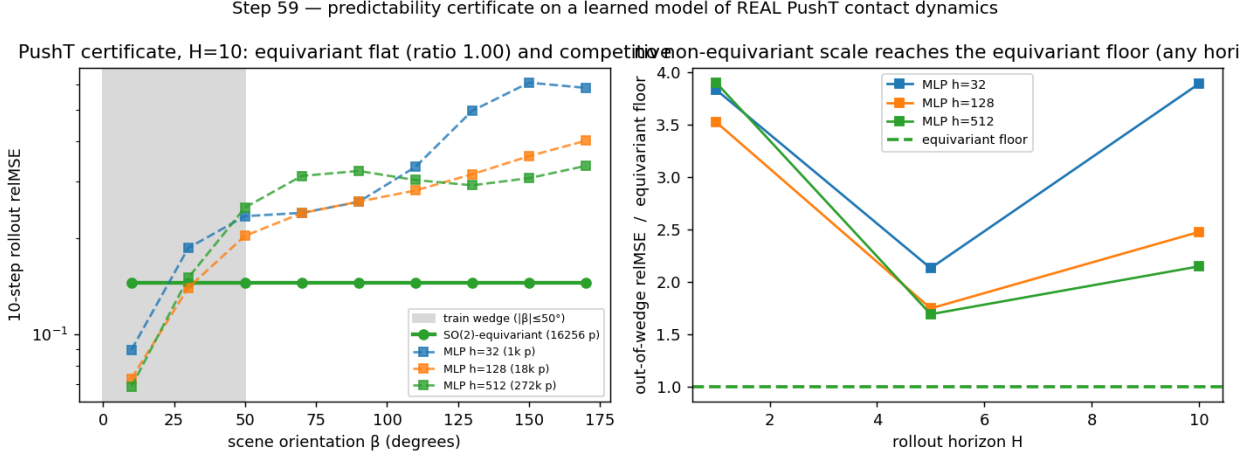


Figure 7: The certificate on real contact dynamics (E5). On PushT (a physics engine we did not author), a *learned* $\text{SO}(2)$ -equivariant world model’s multi-step rollout error is flat over the orbit of scene orientations to the floating-point floor, while a $160\times$ parameter sweep of non-equivariant baselines never reaches the equivariant floor out of distribution.

(E6) The certificate is not $\text{SO}(2)$ -specific, and transfers to pixels at no accuracy cost. It lifts to non-abelian $\text{SO}(3)$ on 3D point clouds (learned equivariant rollout exactly flat, ratio 1.000, with a $7.4\times$ -smaller model than the baseline). On raw rendered pixels (PushT, exact C_4), **frame averaging** (Puny et al. 2022) — a plain CNN/MLP made exactly C_4 -equivariant by a Reynolds average over grid rotations — keeps the exact orbit-flat certificate (ratio 1.000) while being **accuracy-neutral**: it matches or beats an unconstrained CNN on a collapse-robust metric (fraction-of-variance-unexplained ratio 0.68–1.07, mean 0.84, 3 seeds), with a healthier latent and a horizon-stable rollout where the steerable baseline diverges. The prior costs nothing *relative to* the unconstrained CNN — though both remain limited in absolute terms (neither beats predict-the-mean at this scale; §6).

(E7) The certificate becomes task competence. Run through a G -equivariant planner (A5), the prediction certificate becomes an *orbit-invariant control* certificate: closed-loop pose error is flat over the orbit to the floating-point floor (ratio 1.000, 3 seeds) while a $4.3\times$ -larger non-equivariant controller degrades out of the training wedge. The flatness is exact because (A5) makes the planner G -equivariant and the planning cost G -invariant, so the closed-loop trajectory inherits the encoder’s orbit-equivariance — Theorem A under the closed-loop clause.

(E8) The certificate is an epistemic drive: certificate-driven active inference. Because the certified region is *computable* (Algorithm 1), an agent can act to **expand** it — a provable alternative to curiosity’s “reduce prediction error”. On a $\mathbb{Z}_2^7$ compositional world whose action space mixes 7 true generators with noisy high-error distractors, an explorer that maximizes certified-region growth certifies all $2^7=128$ compositions in exactly 7 observations (the generator basis — each generator unlocks an exponential swath, Lemma 1), whereas an error-curiosity explorer is lured by the irreducible-noise distractors and certifies only 1% at the same budget (3 seeds; random $\sim 2\%$). So “expand your certified region” is a **noise-immune** epistemic drive that beats prediction-error curiosity — the classic *noisy-TV* failure mode — for compositional coverage. *Honest scope*: the baseline is *raw-error* curiosity; a sophisticated information-gain agent would also learn to

Frame averaging: exactly flat, accuracy-neutral vs the unconstrained CNN, and horizon-stable (learned pixel latent, C_4)

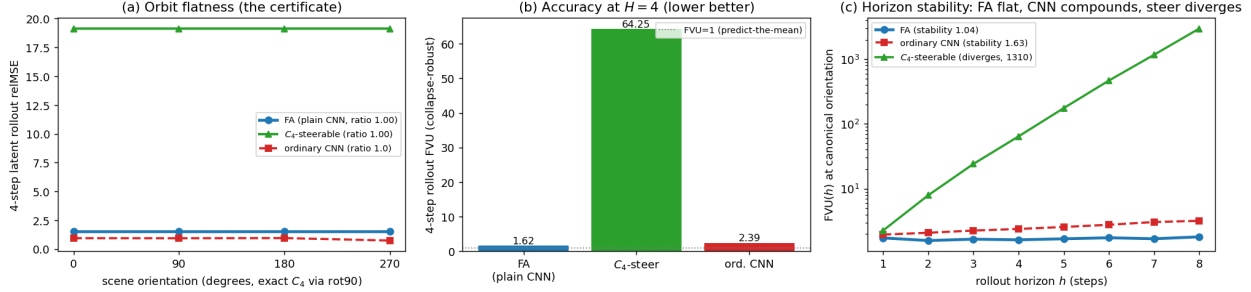


Figure 8: The certificate on raw pixels at no accuracy cost (E6). On rendered PushT frames (exact C_4), frame averaging makes a plain CNN/MLP exactly C_4 -equivariant: the latent rollout is flat over the orbit (left, the certificate), it matches or beats an unconstrained CNN on a collapse-robust accuracy metric (centre), and its rollout error stays bounded over horizon while the steerable baseline diverges (right, log scale).

avoid aleatoric noise. The certificate’s contribution is that it supplies that immunity **for free and provably**, with no noise model — and this is a toy demonstration of the principle, not a benchmark.

5. Related work

Single-shot certified equivariance is our $T=1$ slice. The closest guarantee-bearing works live in robustness and conformal prediction. Equivariant classifiers have *orbit-constant margins* (the decision-boundary gradient norm is preserved across the orbit), giving uniform adversarial certificates (*Orbit-Constant Margins for Equivariant Networks* 2025); invariance-aware randomized smoothing builds *orbit-based* certificates (*Invariance-Aware Randomized Smoothing Certificates* 2022); and Equivariantized Conformal Prediction (eCP) (*Equivariantized Conformal Prediction* 2026) group-averages the nonconformity score — frame averaging for distribution-free coverage. **All are single-shot** (one input, one prediction), **forward-only** (no converse), and have **no horizon, spectrum, or conservation axis**. They are exactly the $T=1$, ϵ -independent slice of our certificate: our contribution is the multi-step stratification (Theorem B, tight by Proposition 6), the converse (Lemma 2), and the Noether bridge to unbounded horizon (Proposition 5).

Learned conservation laws. Noether Networks (Alet et al. 2021) meta-learn conserved quantities inside the prediction loop, and Noether’s Razor (*Noether’s Razor* 2024) learns symmetries-as-conserved-quantities by Bayesian model selection. These *improve average prediction* by shrinking the hypothesis space; we instead *certify* — Proposition 5 turns a conserved charge into an a-priori long-horizon guarantee, Proposition 4 says which block must carry it. Guarantee versus average accuracy.

Jacobian-regularized world models. “Towards Unraveling and Improving Generalization in World Models” (*Towards Unraveling and Improving Generalization in World Models* 2025) penalizes the latent-transition Jacobian to damp rollout error propagation — a heuristic for robustness. Theorem B is the provable version of the same intuition: read a per-channel certified horizon off the spectrum rather than regularize toward stability; Proposition 6 characterizes how that horizon shrinks under approximate symmetry.

Equivariant predictors and latent-geometry priors. Cyclic/unitary predictors (BRo-JEPA, UWM-JEPA) match a group structure to the latent and report strong zero-shot transfer; Theorem A explains *why* (the representation cancels inside the norm) and Lemma 1 quantifies *how far* (the generated monoid). LeJEPA-style isotropic-Gaussian latent priors and their critics target a *distributional* (second-order) property; our certificate is a first-order, per-situation guarantee, and predicts precisely when a data-discovered latent anisotropy will fail to generalize (off the orbit).

Predictability horizons. The $T(\epsilon) \sim \log(1/\epsilon)/\lambda$ law is classical for dynamical systems (Lyapunov; numerical weather prediction); that the *local* spectrum governs the *asymptotic* rate is the Oseledets multiplicative

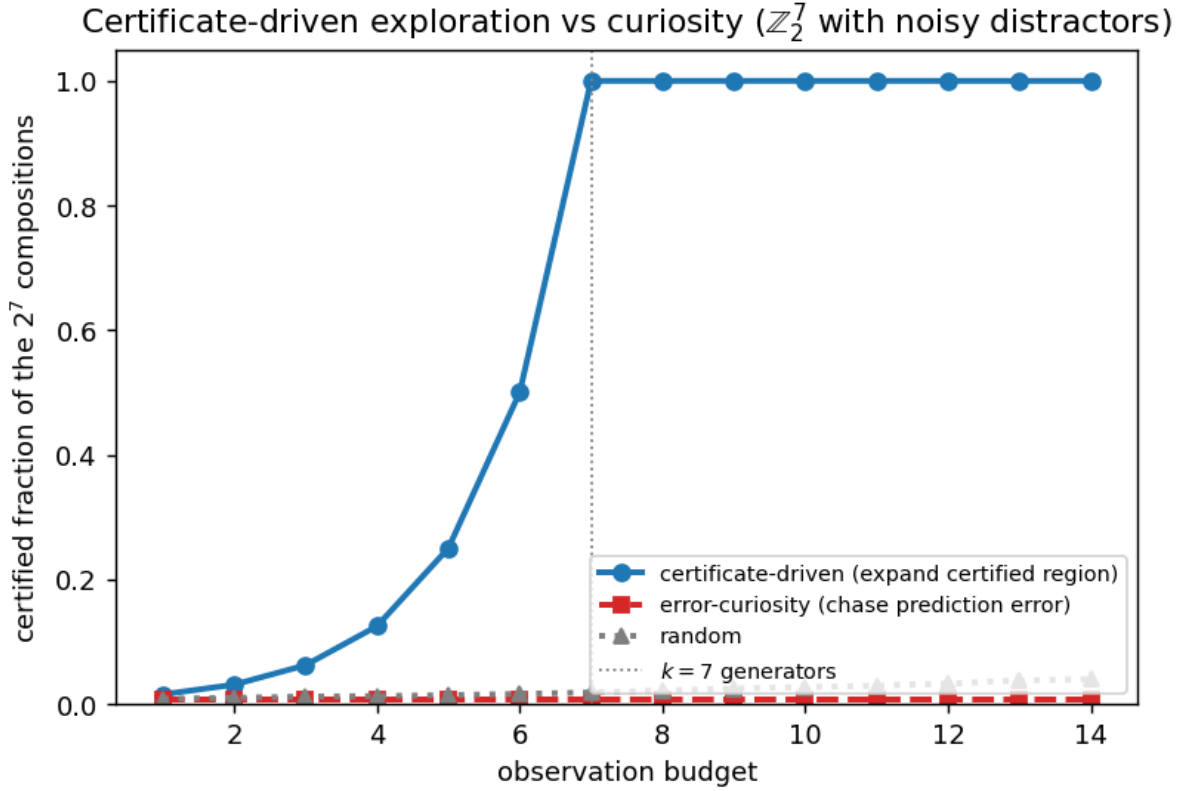


Figure 9: Certificate-driven active inference (E8). On a $\mathbb{Z}_2^7$ compositional world with noisy distractor actions, an explorer that maximizes the *certified region* certifies all 128 compositions in exactly 7 observations (the generator basis), while a prediction-error-curiosity explorer is lured by the high-error distractors and certifies almost nothing at the same budget.

ergodic theorem (Oseledets 1968), and on uniformly hyperbolic systems the shadowing lemma (Pilyugin 1999) bounds a perturbed model’s forecast-horizon floor (it controls orbit error, not the exponent). Our contribution is to (i) prove the law *tight* for a *learned* latent world model and **measure it on a learned model of genuinely chaotic dynamics** (Lorenz, E2 — the learned model’s exponent matches the true one to 1–8%); (ii) **characterize when it lifts** (Proposition 7: the non-degenerate/degenerate dichotomy, synthesizing Oseledets for the rate with shadowing for the floor); (iii) tie its unbounded-horizon subspace to the group-invariant subspace (Noether hinge); and (iv) make the forward direction’s converse a theorem (Lemma 2).

6. Limitations

- **Exactness needs a genuine dynamical symmetry (A3).** The certificate is *exact* where G is a symmetry of the dynamics (orbital/conservative systems, free space, idealized manipulation) and *gracefully approximate* elsewhere, with a degradation that is now bounded on both sides (Theorem B and Proposition 6) and measured (E3).
- **Tight in form, not in prefactor.** Proposition 6 makes the horizon’s $\log(1/\epsilon)/\lambda$ *form* tight, but the constants c_j are not estimated and the isotypic refinement is asserted, not separately measured.
- **The Noether hinge’s forward direction is proved; its hypotheses are measured.** Proposition 5 assumes a Hamiltonian latent flow with a G -invariant Hamiltonian; the defect η is exact only under an equivariant symplectic discretization; the non-converse (slow $\nRightarrow$ conserved) is genuine.
- **The hinge is validated on constructed teachers, not shown to *emerge*.** The Noether experiments (E2 and the containment results) use *constructed* equivariant teachers, so the “slow = invariant” coincidence is partly built in. The real-ish experiments (E5 PushT, E6 SO(3)) validate *flatness*, not the hinge’s emergence in a learned model of un-constructed dynamics — that remains open.
- **The horizon axis lifts to learned models of real *chaotic* dynamics, but the lift is empirical, and is vacuous on near-neutral dynamics (scope, not a bug).** E2 now validates the certified-horizon law on three levels: a synthetic constant-spectrum latent, analytically (Proposition 6, step65), and — the decisive case — a *learned* model of the genuinely chaotic **Lorenz** system, where the staircase is linear ($R^2=0.975\text{--}0.995$) and the learned model’s Lyapunov exponent matches the true λ_1 to 1–8%. Proposition 7 explains the regime structure: the horizon axis is informative iff the dynamics is spectrally non-degenerate ($\lambda_1 > 0$). The **PushT interior is the degenerate branch** ($\lambda_1 \approx 0$, near-neutral): a learned-PushT probe (**experiments/step67**) finds the local Jacobian spectrum does *not* predict the multi-step rollout ($R^2 \approx 0.02$) — *predicted* by Proposition 7(b), not a defect of the certificate. Two honest caveats remain. (i) The lift is **empirical, not theorem-backed**: Oseledets gives the rate rigorously, but shadowing bounds only the forecast-horizon *floor* and does **not** transfer the Lyapunov exponent (exponents are upper-semicontinuous under C^1 perturbation), and classical shadowing does not even formally cover singular-hyperbolic Lorenz; that the learned model reproduces λ_1 is the *measured* finding, and we verify C^1 -closeness only via one-step error (an L^2 proxy). (ii) The *configuration axis* (orbit-flatness, ratio 1.000) is the part that holds on real PushT at the contact level; the horizon axis’s real-dynamics evidence is Lorenz, a 3D ODE, not a contact simulator.
- **Scale and modality.** All experiments are 1–2-GPU. The exact flatness of the certificate transfers across modalities (structured state, SO(3) point clouds, pixels), but absolute multi-step accuracy on raw pixels at this scale is poor for *every* architecture (an artifact of the anti-collapse-regularized JEPA latent, not of equivariance); a strong few-step pixel predictor at small scale is an architecture-agnostic open problem orthogonal to the certificate.

7. Conclusion

An equivariant world model can certify, a priori and without retraining, which situations it will handle and **for how many steps** — and the horizon is tight. The certified region is the coarse-invariant-slow-low-composition corner; its boundary is the predictor spectrum; its unbounded edge is the conserved/invariant

subspace (Noether); and the whole region is reachable only with structure (the converse). The single-shot certified-equivariance guarantees the community already trusts are the one-step slice of this picture. *Scale buys interpolation; structure buys a certified horizon.*

Companion note. This paper is a focused extraction of a broader certified-world-models program; the full experiment suite, the closed-loop clause, the discovery/generation re-framing, and the complete proofs appear in a companion manuscript (omitted here for anonymity).

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